AI TOOL PROMPT ENGINEERING REPORT

Presented to the

Elective 3 Course

Ateneo de Davao University

E. Jacinto St., Davao City

In Partial Fulfillment of the Requirements for:

GE-4120: 21st Century IT Skills

Submitted by:

Legario, Alfred Glic B.

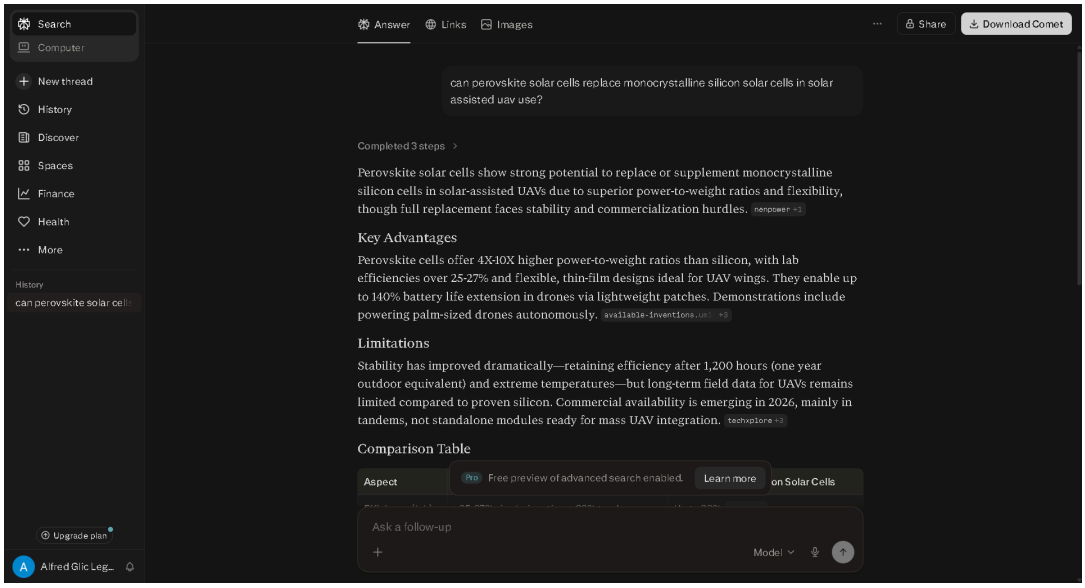
Submitted to:

Prof. Gabriel Quibod

April 2026

AI Tool Overview and Original Prompt

As engineering students, we are always on the look out for modern solutions to contemporary problems. Currently, we are developing a study on the Perovskite that could possibly challenge the industry standard, Monocrystalline Silicon, hence this prompt. The chosen AI tool was the Perplexity AI, as it is an AI-powered, search and information synthesis search engine, best known for having surface information on a topic.

Prompt	“can perovskite solar cells replace monocrystalline silicon solar cells in solar assisted uav use?”
Result	

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limited compared to proven silicon. Commercial availability is emerging in 2026, mainly in tandems, not standalone modules ready for mass UAV integration. techxplore +3

Comparison Table

Aspect	Perovskite Solar Cells	Monocrystalline Silicon Solar Cells
Efficiency (lab)	>25-27% single-junction; >33% tandem nenpower +1	Up to 29% nenpower
Power/Weight Ratio	4-10X higher (>400 W/kg potential) available-inventions.um +1	Lower (~20 W/kg) tongwei
Flexibility	Ultra-flexible, lightweight patches available-inventions.um +1	Flexible variants exist but heavier (e.g., 645 g/m² corrugated) bohrium +1
Stability	1,200+ hours light; thermal cycles techxplore +1	Decades proven techxplore
UAV Suitability	Proven in prototypes for extended flight available-inventions.um +1	Used in flexible forms for endurance bohrium +1

Feasibility Outlook

Perovskites are viable for near-term UAV augmentation, boosting endurance without much added weight, but won't fully replace silicon until 2027+ mass production and 25-year warranties. Tandem hybrids could bridge the gap for optimal performance. greenfueljournal +3

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20 sources

Follow-ups

- Build a comparison dashboard for perovskite vs monocrystalline silicon solar cells in UAVs showing power-to-weight, efficiency, stability, and flight range extension metrics — Computer
- Create a PDF report comparing perovskite and silicon solar cells for UAV applications with performance tables, stability data, and key research summaries — Computer
- What are main stability challenges for perovskites in UAVs
- How do flexible perovskite cells integrate into UAV wings
- Perovskite-silicon tandem cells benefits for drones

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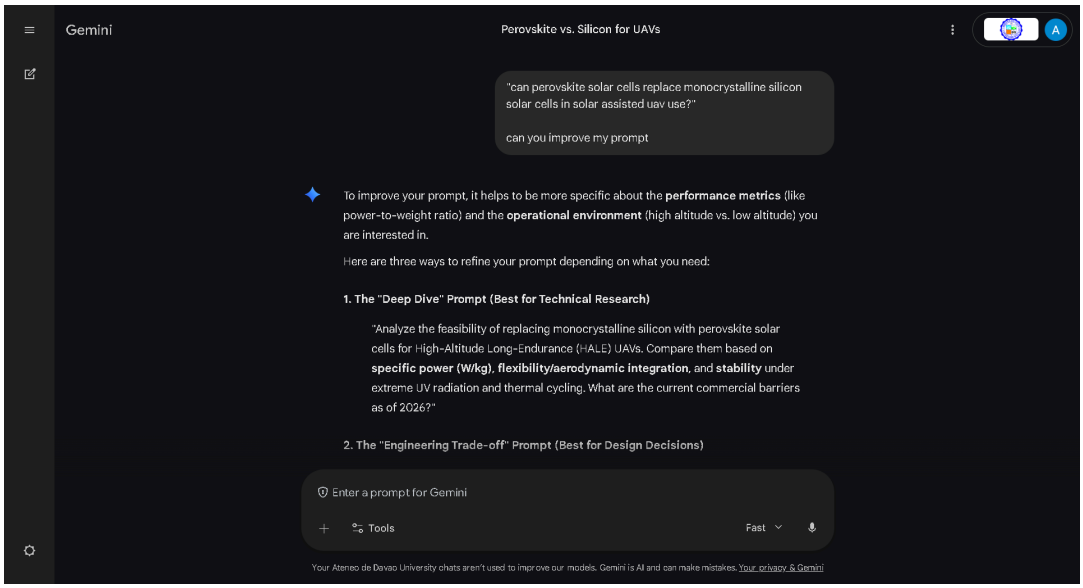
AI Prompt Consultation

The chosen LLM to generate the AI-Assisted prompt was Google Gemini, and it was tasked to improve the raw prompt for comparison. Given that I also used Google Gemini for consultation, I made sure to make a new chat for a clean slate.

Prompt	"can perovskite solar cells replace monocrystalline silicon solar cells in solar assisted uav use? can you improve my prompt”
Result	

Gemini

Perovskite vs. Silicon for UAVs

as of 2026?"

2. The "Engineering Trade-off" Prompt (Best for Design Decisions)

"Compare monocrystalline silicon and perovskite solar cells for a solar-powered UAV with a 5-meter wingspan. Specifically, look at how the **weight savings** of perovskite might offset its shorter **operational lifespan** compared to silicon. Which technology currently offers the best 'flight-hours per dollar'?"

3. The "Future Outlook" Prompt (Best for Trend Analysis)

"Given the 2026 milestones in **perovskite-silicon tandem cells**, should UAV manufacturers switch entirely to thin-film perovskite, or is a tandem structure more viable for maximizing endurance? Focus on the latest efficiency records and moisture-resistance breakthroughs."

Comparison Summary: Perovskite vs. Silicon for UAVs

If you're looking for a quick answer on whether they *can* replace silicon right now, here is the state of the tech as of 2026:

Feature	Monocrystalline Silicon	Perovskite Solar Cells	Impact on UAVs

Enter a prompt for Gemini

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Gemini

Perovskite vs. Silicon for UAVs

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Comparison Summary: Perovskite vs. Silicon for UAVs

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Feature	Monocrystalline Silicon	Perovskite Solar Cells	Impact on UAVs
Weight	Relatively heavy/rigid	**Ultra-lightweight** (thin-film)	Perovskite allows fo much higher paylo capacity.
Efficiency	~20–24% (Commercial)	**25–27%** (Single-junction)	Perovskite provide more power per cr wing area.
Flexibility	Brittle; requires flat surfaces	**Highly flexible**	Perovskite can be "wrapped" around curved wing foils.

Enter a prompt for Gemini

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Gemini

Perovskite vs. Silicon for UAVs

Heat Tolerance	Drops efficiency as it gets hot	**Maintains efficiency**	Perovskite perform better in direct, int high-altitude sun.

Export to Sheets

The Verdict

Perovskite is currently the "**performance king**" for niche, high-end UAVs where every gram matters (like record-breaking endurance drones). However, for commercial or consumer drones, **monocrystalline silicon** remains the standard because it doesn't degrade when it gets rained on or sits in a humid warehouse.

The biggest trend in 2026 is actually **Tandem Cells**, which layer perovskite *on top* of silicon to reach efficiencies over **35%**, offering a "best of both worlds" solution for high-end aerospace.

Which specific type of UAV are you looking at—a small hobbyist drone or a large high-altitude glider?

Enter a prompt for Gemini

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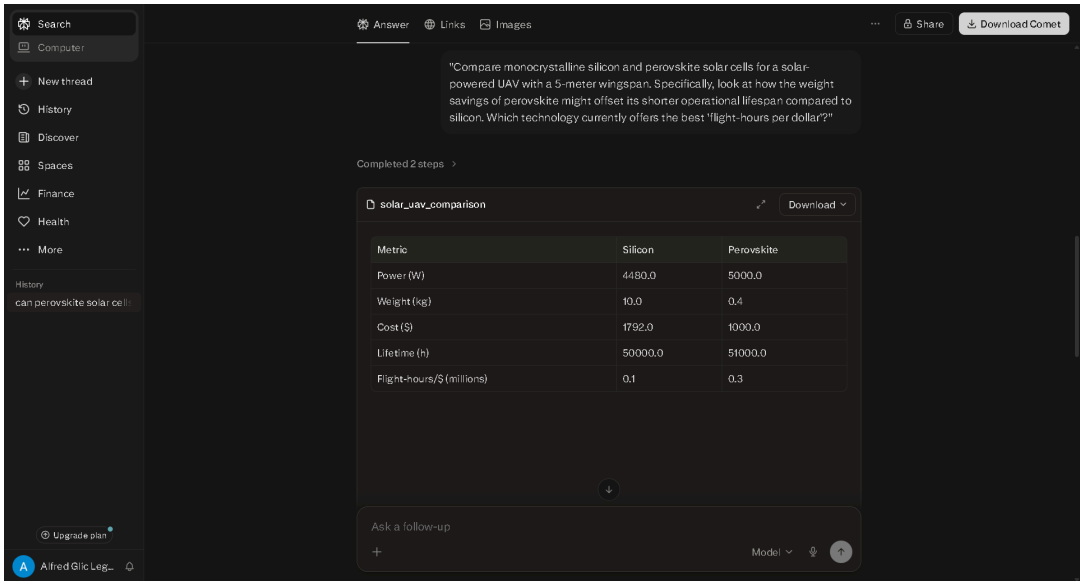
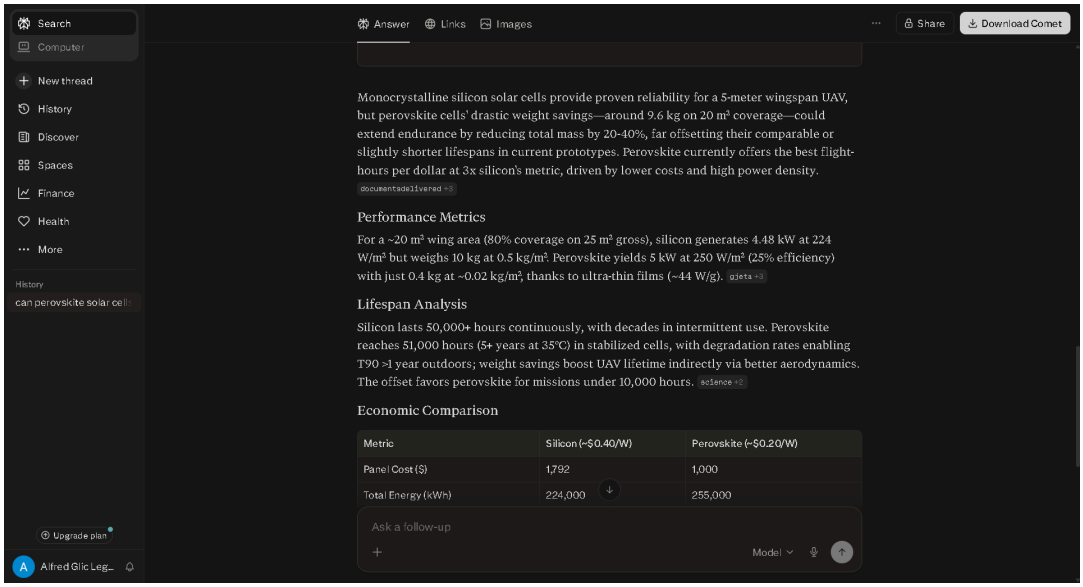
AI-Assisted Prompt

Google Gemini gave three (3) improved prompts, I went with the number two (2) suggestion, as this was the closest to the earlier prompt, both addressing the design of the UAV.

Prompt	"Compare monocrystalline silicon and perovskite solar cells for a solar-powered UAV with a 5-meter wingspan. Specifically, look at how the weight savings of perovskite might offset its shorter operational lifespan compared to silicon. Which technology currently offers the best 'flight-hours per dollar'?"
Result	 

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Economic Comparison

Metric	Silicon (~$0.40/W)	Perovskite (~$0.20/W)
Panel Cost ($)	1,792	1,000
Total Energy (kWh)	224,000	255,000
Flight-hours/$ (millions)	0.1	0.3

Perovskite wins on $/Wh due to half the cost and higher output, assuming equalized lifetimes; real prototypes confirm extended flights. 6msolar +1

Best Choice

Perovskite delivers superior flight-hours per dollar today for cost-sensitive UAVs, with weight benefits amplifying endurance despite minor lifespan gaps—ideal for prototypes scaling to 5m spans. Silicon suits ultra-long missions needing guaranteed 20+ year durability. oaspublish +2

11 sources

Follow-ups

- Why perovskite solar UAV projects have repeatedly failed commercialization despite lab weight advantages over silicon — key engineering hurdles and latest breakthroughs
- What stability improvements make perovs viable for UAVs

Deep research

Ask a follow-up

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- What stability improvements make perovskite viable for UAVs
- Power output estimates for perovskite on 5m wingspan UAV
- Cost comparison of perovskite vs monocrystalline for UAVs
- Real-world flight endurance tests with perovskite drones

Deep research

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Output Comparison

Comparing the two AI outputs shows how changing a prompt directly changes the depth and usefulness of the results. The original prompt about perovskite and silicon for UAVs was broad, which led to a basic overview of pros, cons, and general ideas. While helpful, that first response stayed general, focusing on lab ratings and future market hurdles. It did not offer the technical depth needed for actual engineering decisions, providing instead a high-level summary that felt more like an introductory article than a technical report.

In contrast, the improved prompt added specific details: a 5-meter wingspan and a "flight-hours per dollar" target. This shift forced the AI to move away from general talk and toward real engineering math. The resulting data was much more practical, providing calculated power totals of 224 W/m² for silicon versus 250 W/m² for perovskite, along with a weight savings of 9.6 kg. By looking at how the lighter weight of perovskite makes up for its shorter life, the second output gave a clear economic answer: perovskite is a better value for long-distance flight. This type of analysis is vital for making real-world choices in aerospace design where every gram counts.

The main differences are in the details and the accuracy. The broad prompt led to general guesses, while the improved version triggered actual data and math. The second output was much more useful because it provided the kind of solid information, such as total aircraft weight reduction, that a designer could actually use. This shows that better prompts depend on clear limits and specific goals. Vague questions get opinions, but adding details like size and specific trade-offs gets evidence-based answers. Moving forward, the most effective technical prompts will be those that set a clear way to measure success. This process of sharpening a prompt is the best way to get the most useful and high-quality work out of an AI. This method ensures the output is both relevant and reliable for real-world tasks, helping users solve complex problems with much greater certainty.